02 BUS SELECT, GROUND CANCEL BUFFER, SUBGROUP

File: group_sheets/02_bus_select.kicad_sch

03 SUBGROUP

File: group_sheets/03_subgroup.kicad_sch

04 SUBGROUP, AUX MASTER

File: group_sheets/04_subgroup.kicad_sch

05 AUX MASTER, STEREO RETURN

File: group_sheets/05_aux_master.kicad_sch

06 STEREO RETURN

File: group_sheets/06_stereo_return.kicad_sch

07 STEREO RETURN

File: group_sheets/07_stereo_return.kicad_sch

08 STEREO RETURN 2

File: group_sheets/08_stereo_return_2.kicad_sch

09 STEREO RETURN 2

File: group_sheets/09_stereo_return_2.kicad_sch

10 STEREO RETURN 2, JACK BOARD LINK, JACK BOARD LINK 2

File: group_sheets/10_stereo_return_2.kicad_sch

11 POWER AND GROUNDS, RIBBON

File: group_sheets/11_power_and_grounds.kicad_sch

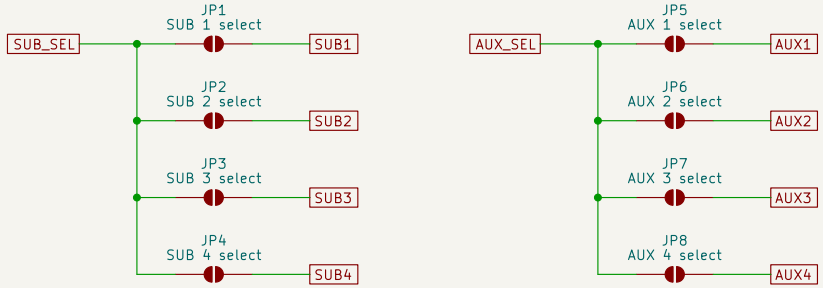
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)

Sheet: /
File: group.kicad_sch

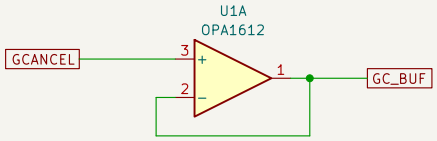
Title: Group module — goa console (sub + return + aux master)

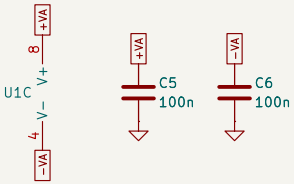
Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 1/11

BUS SELECT close ONE SUB jumper and ONE AUX jumper: group n sums SUB n and AUX n (all open = summer idle)

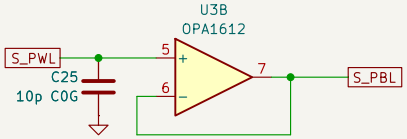


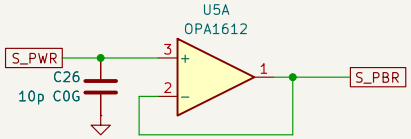
GROUND CANCEL BUFFER (local, for this module's re-inverters; summing amps use the raw GCANCEL line)

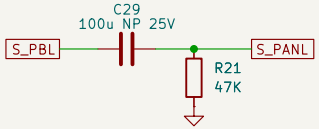


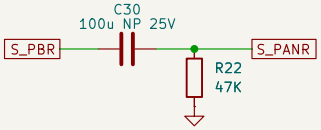


SUBGROUP: sum (selected SUB bus), re-invert, insert, fader + post-amp, DRV134 out, pan to MIX, AFL [spec 12]









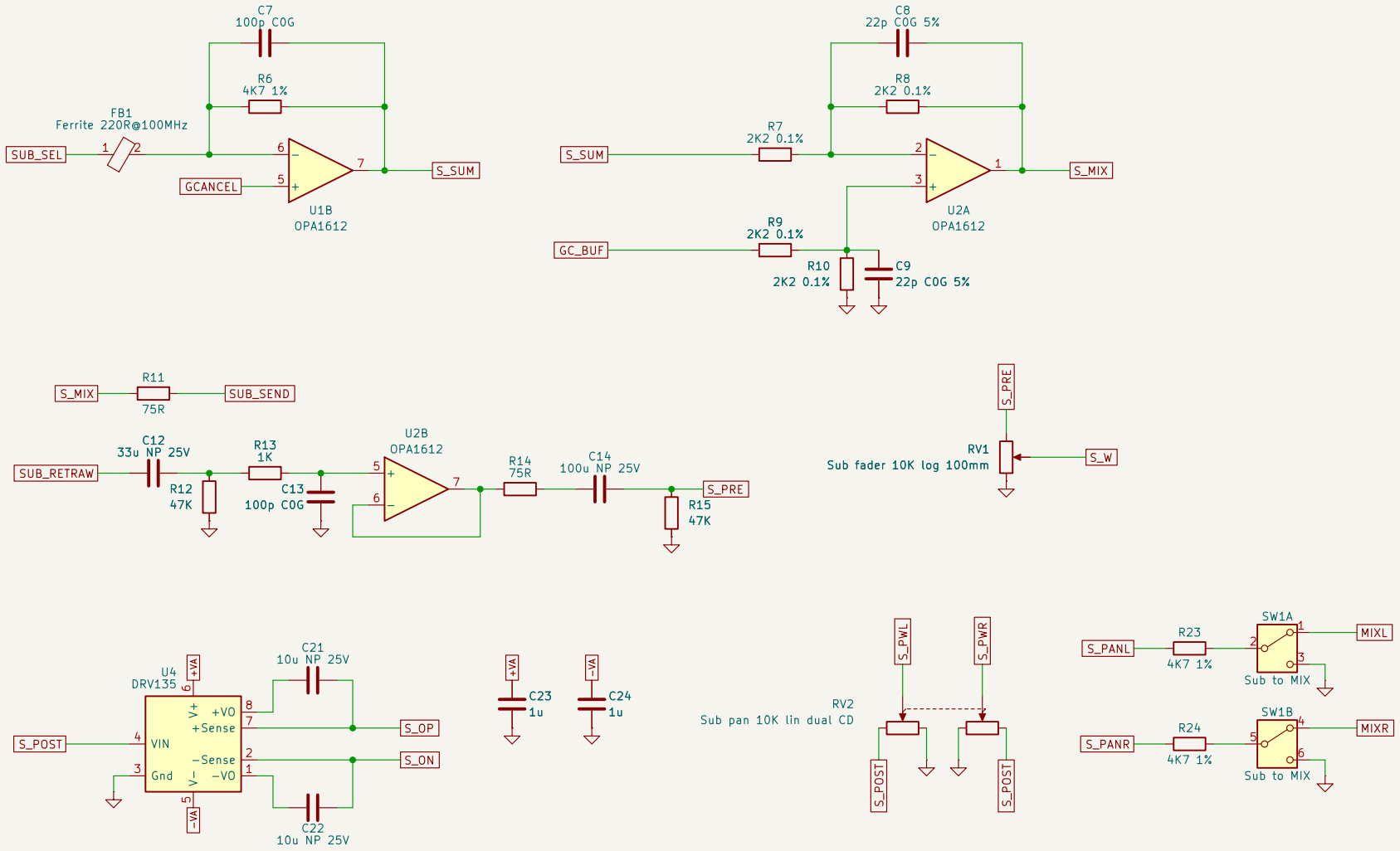
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /02 BUS SELECT, GROUND CANCEL BUFFER, SUBGROUP/
File: 02_bus_select.kicad_sch

Title: BUS SELECT, GROUND CANCEL BUFFER, SUBGROUP

Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 2/11

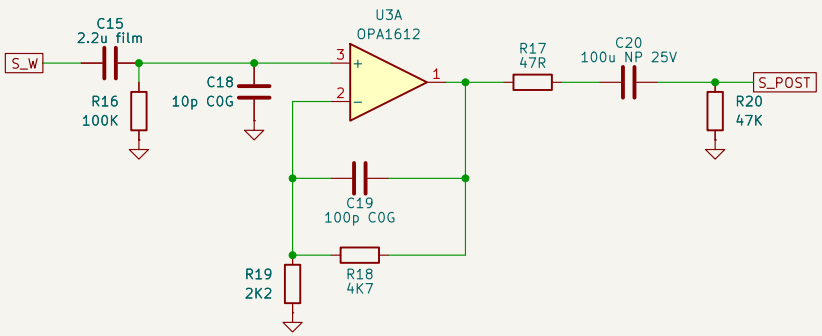
SUBGROUP: sum (selected SUB bus), re-invert, insert, fader + post-amp, DRV134 out, pan to MIX, AFL (cont.)



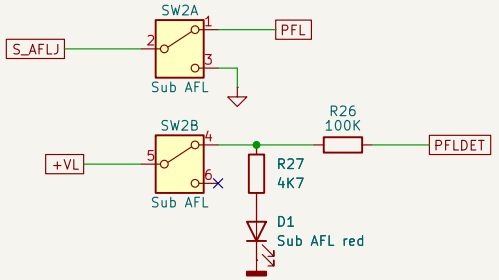
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

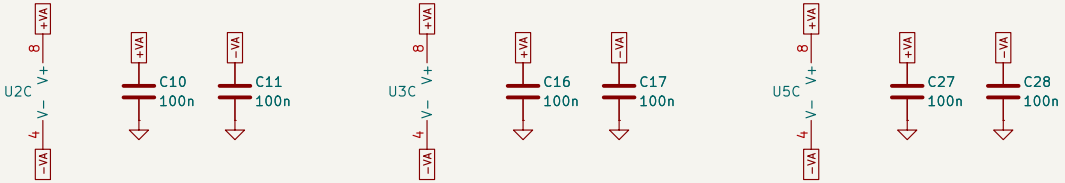
Sheet: /03 SUBGROUP/		
File: 03_subgroup.kicad_sch		
Title: SUBGROUP		
Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 3/11

SUBGROUP: sum (selected SUB bus), re-invert, insert, fader + post-amp, DRV134 out, pan to MIX, AFL (cont.)

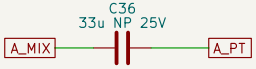


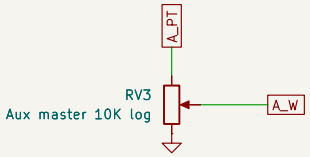


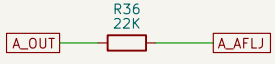


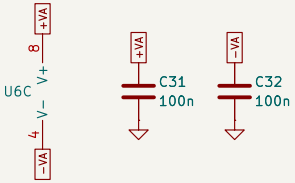


AUX MASTER: OPA1612 sum (selected AUX bus, 22K || 100p) + re-invert, level, buffer, DRV134 out, AFL [spec 13]









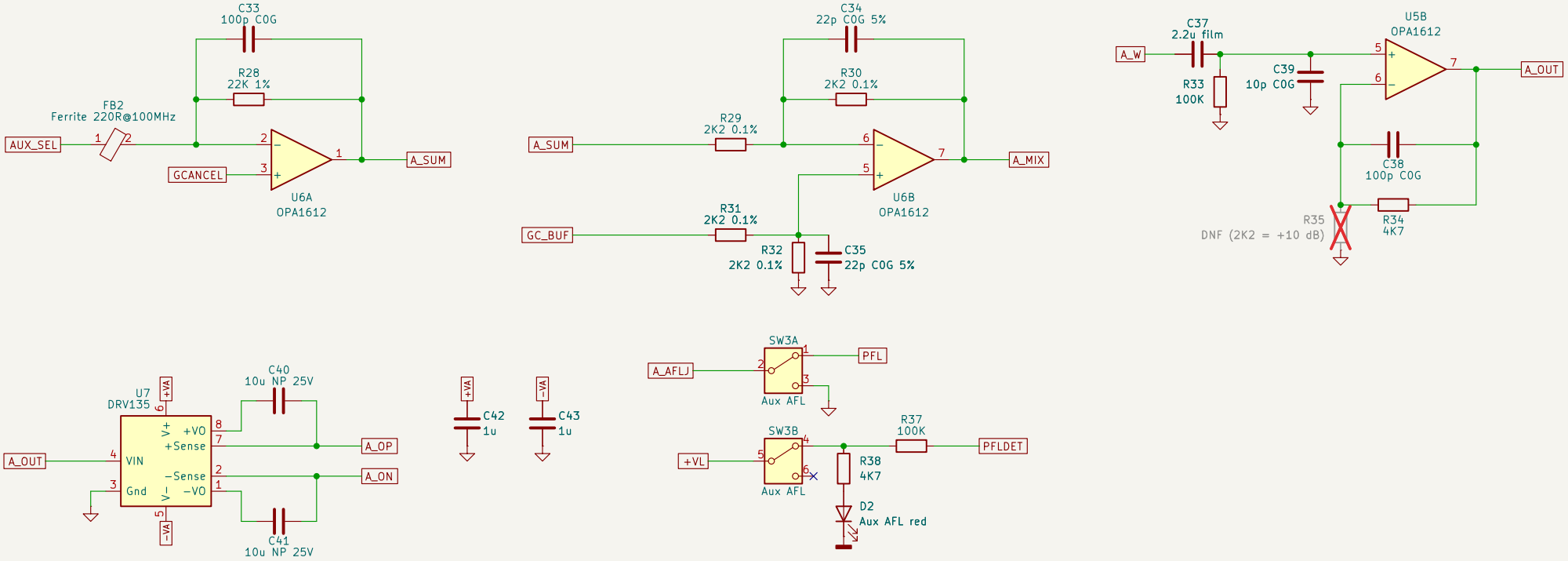
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /04 SUBGROUP, AUX MASTER/
File: 04_subgroup.kicad_sch

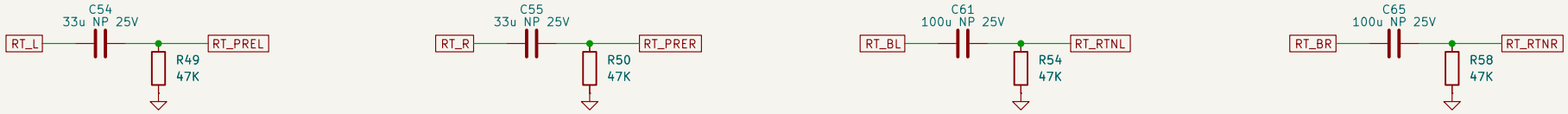
Title: SUBGROUP, AUX MASTER

Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 4/11

AUX MASTER: OPA1612 sum (selected AUX bus, 22K || 100p) + re-invert, level, buffer, DRV134 out, AFL (cont.)



STEREO RETURN: receivers (R normalled to L), level, buffers, MIX / 1-2 / 3-4 onto the ribbon, PFL [spec 16]



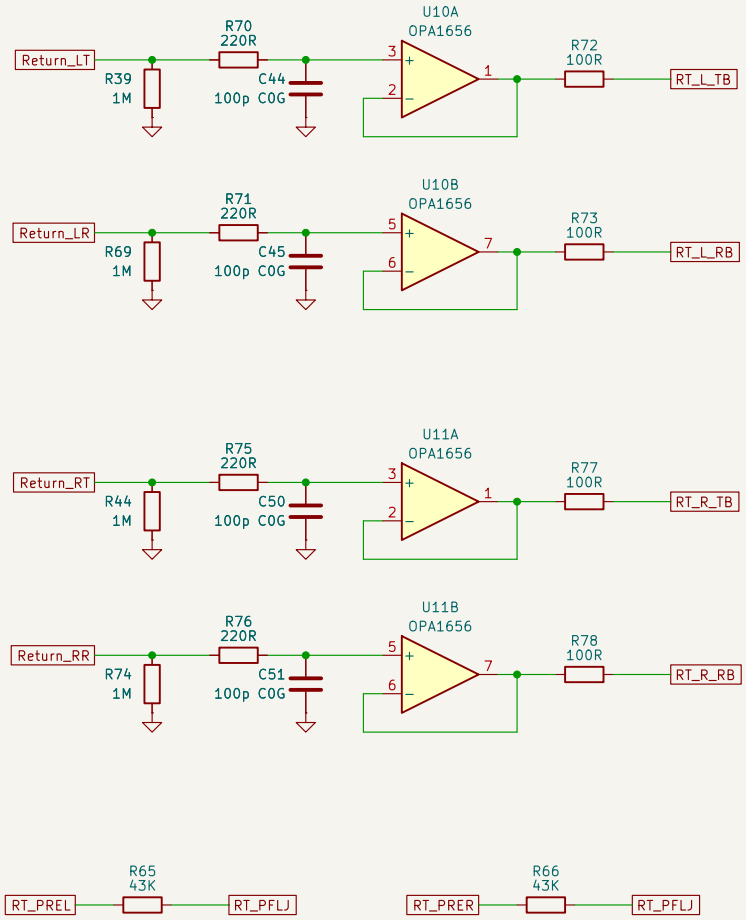
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

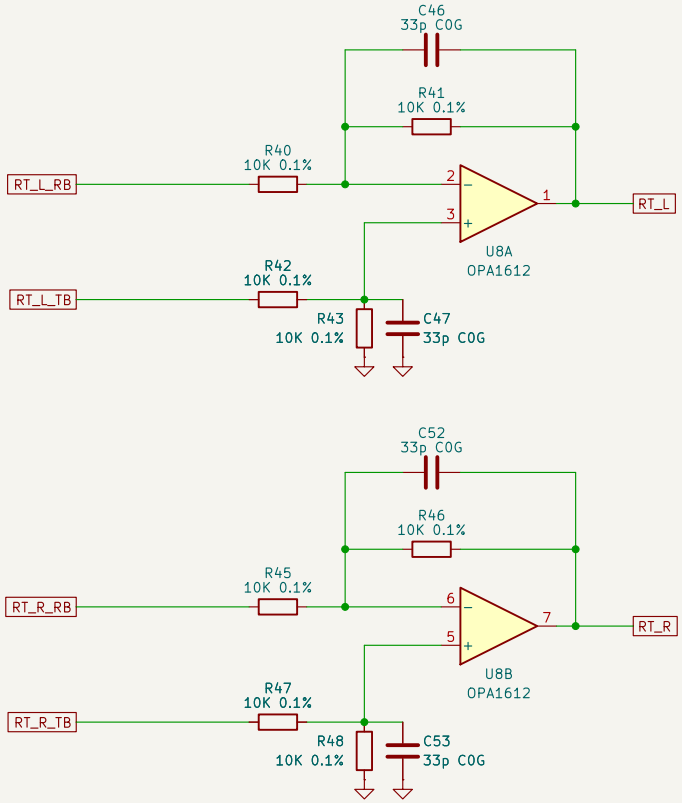
Sheet: /05 AUX MASTER, STEREO RETURN/
File: 05_aux_master.kicad_sch

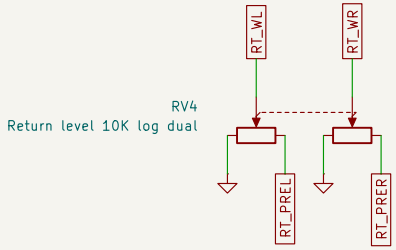
Title: AUX MASTER, STEREO RETURN

Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 5/11

STEREO RETURN: receivers (R normalled to L), level, buffers, MIX / 1–2 / 3–4 onto the ribbon, PFL (cont.)







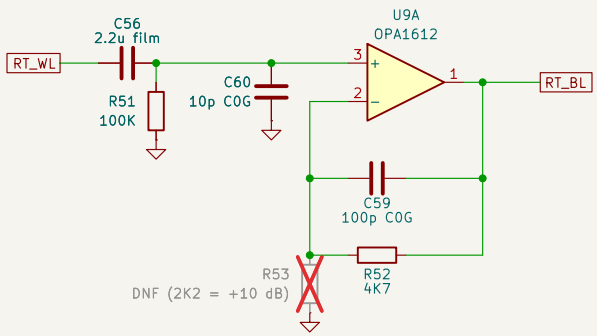
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

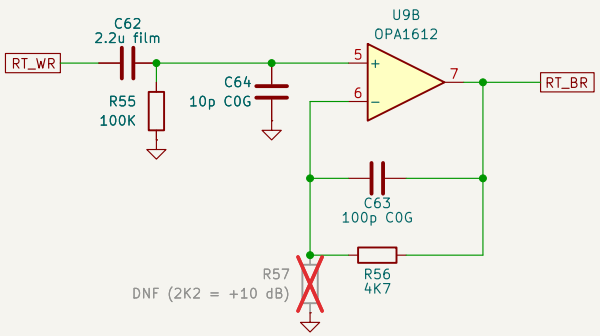
Sheet: /06 STEREO RETURN/
File: 06_stereo_return.kicad_sch

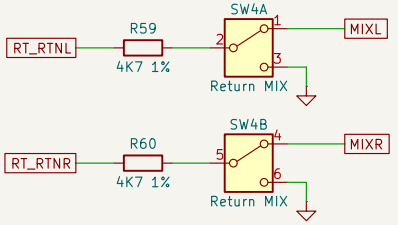
Title: STEREO RETURN

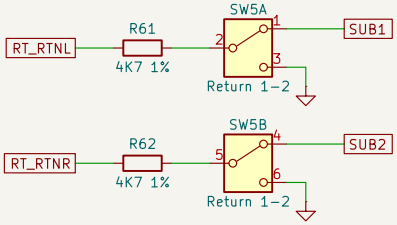
Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 6/11

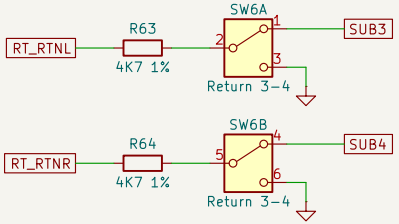
STEREO RETURN: receivers (R normalled to L), level, buffers, MIX / 1–2 / 3–4 onto the ribbon, PFL (cont.)

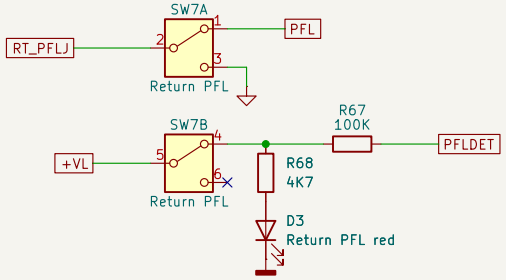


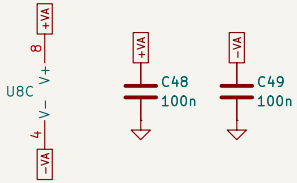


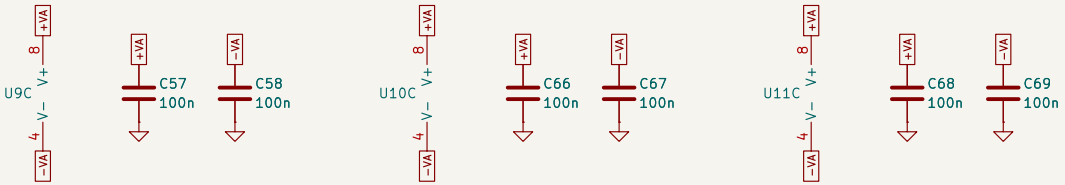












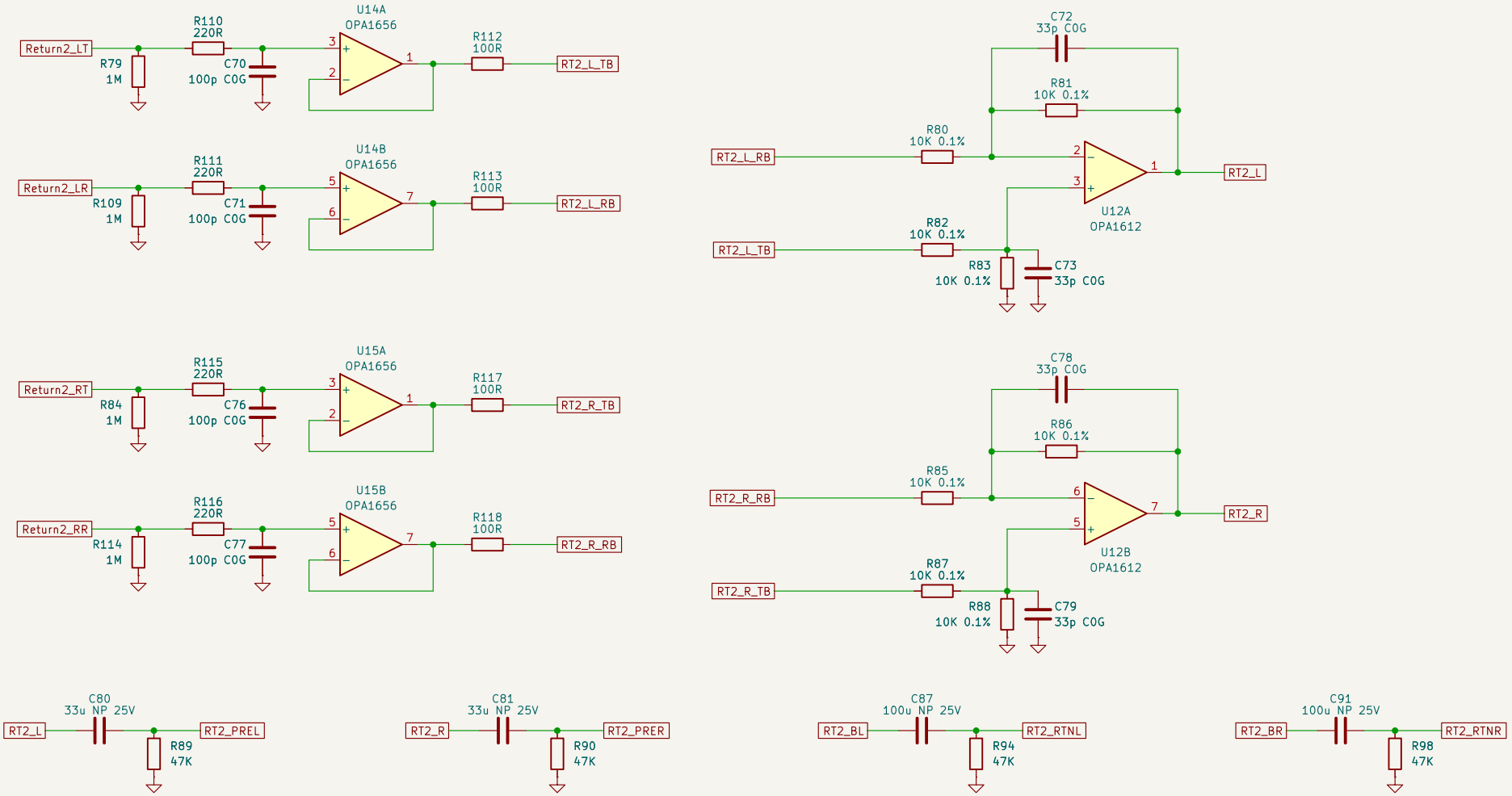
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /07 STEREO RETURN/
File: 07_stereo_return.kicad_sch

Title: STEREO RETURN

Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 7/11

STEREO RETURN 2: as return 1 (receivers, level, buffers, MIX / 1–2 / 3–4, PFL) [spec 16]



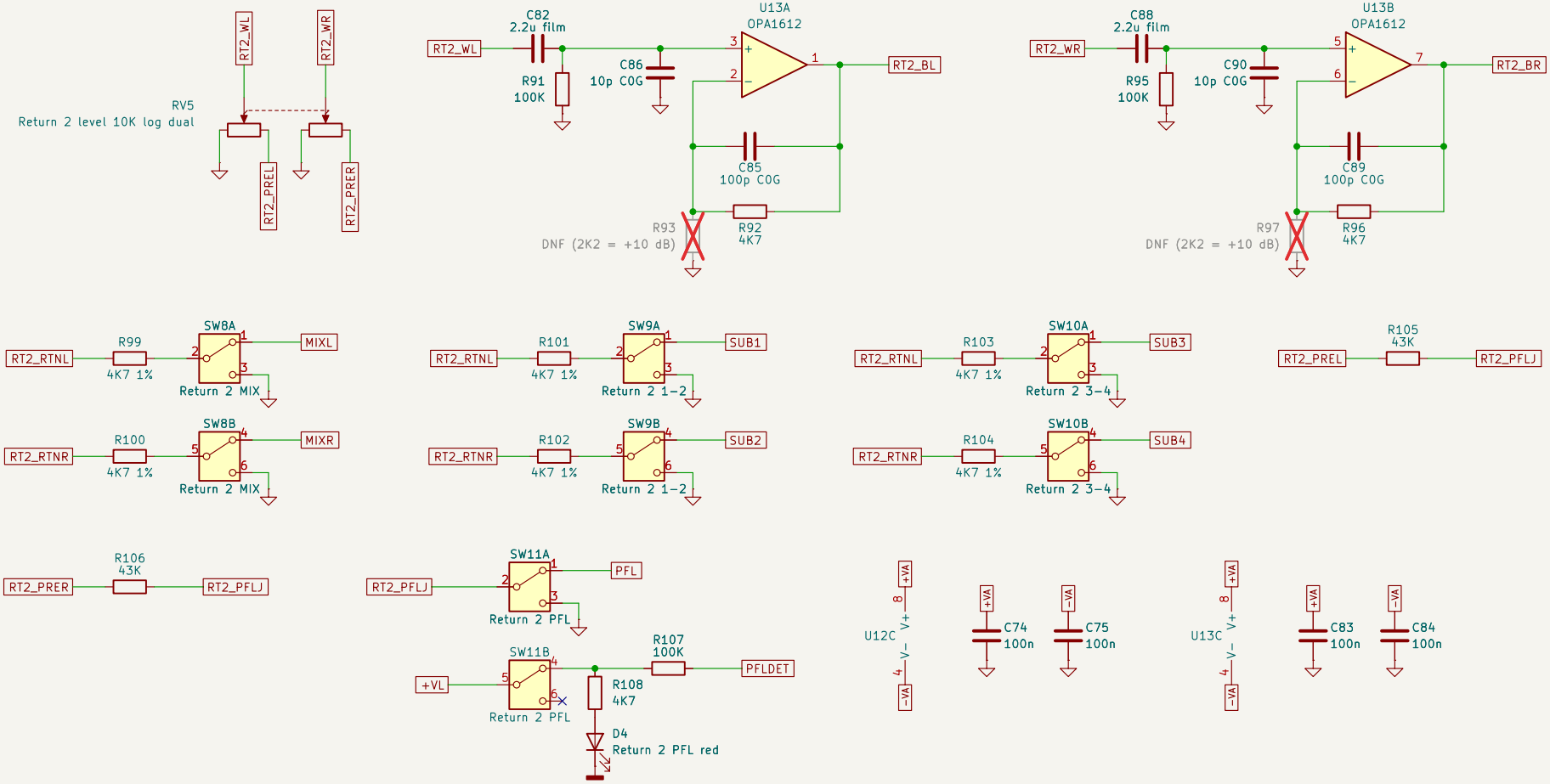
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /08 STEREO RETURN 2/
File: 08_stereo_return_2.kicad_sch

Title: STEREO RETURN 2

Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 8/11

STEREO RETURN 2: as return 1 (receivers, level, buffers, MIX / 1–2 / 3–4, PFL) (cont.)



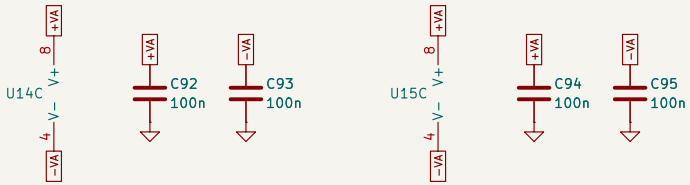
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /09 STEREO RETURN 2/
File: 09_stereo_return_2.kicad_sch

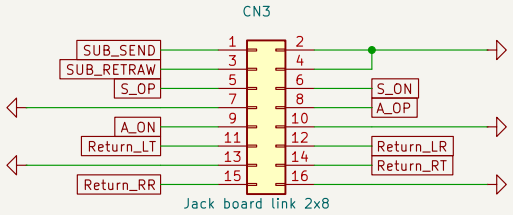
Title: STEREO RETURN 2

Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 9/11

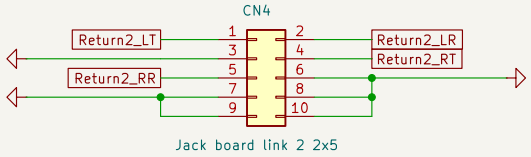
STEREO RETURN 2: as return 1 (receivers, level, buffers, MIX / 1–2 / 3–4, PFL) (cont.)



JACK BOARD LINK 2x8 IDC ribbon to the group jack board (sub, insert, aux, return 1); bottom side, as the strip's CN3



JACK BOARD LINK 2 2x5 IDC ribbon to the group jack board (return 2); bottom side, beside the first



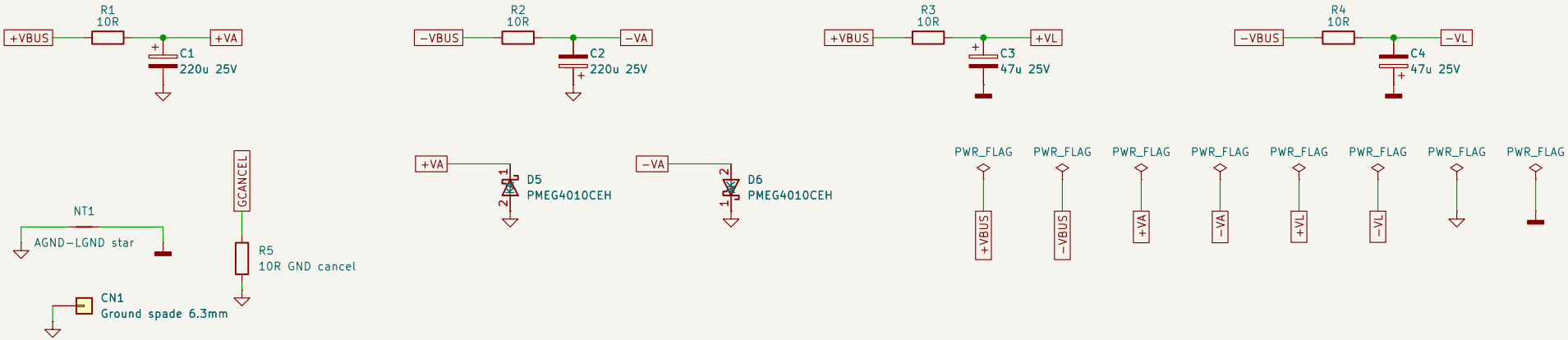
Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module — goa console (sub + return + aux master)

Sheet: /10 STEREO RETURN 2, JACK BOARD LINK, JACK BOARD LINK 2/
File: 10_stereo_return_2.kicad_sch

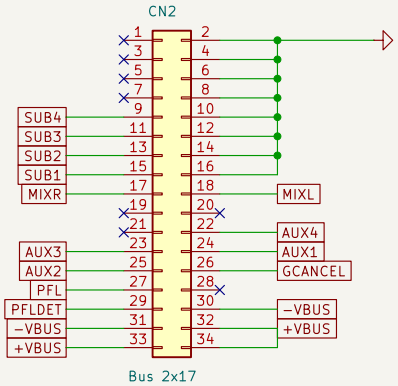
Title: STEREO RETURN 2, JACK BOARD LINK, JACK BOARD LINK 2

Size: A4	Date: 2026-10-19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 10/11

POWER AND GROUNDS (as a strip: 10R + 220u per audio rail, 10R + 47u logic rails, spade, 10R ground cancel)



RIBBON one 34−pin header, Delta pinout (as a strip); AUX 5−6 are summed on the master module



Four identical boards; close one SUB and one AUX select jumper per board
Generated by tools/gen_group.py (master_lib.py; master spec 12, 13, 16)
Group module − goa console (sub + return + aux master)

Sheet: /11 POWER AND GROUNDS, RIBBON/		
File: 11_power_and_grounds.kicad_sch		
Title: POWER AND GROUNDS, RIBBON		
Size: A4	Date: 2026−10−19	**Rev: 0.2**
KiCad E.D.A. 10.0.7		Id: 11/11